Computer Architecture Homework 4

1)

a)Time taken by the stage which takes the maximum time for performing operations=145ps.

Register delay = 12 ps

Therefore, the total time taken by the stage which takes the maximum time = 145 ps + 12 ps = 157 ps

The number of cycles for executing 1000 instructions = 6 (for first instruction) + 999 (For the remaining 1000 – 1 = 999 instructions) = 1005 cycles.

Time taken by 1000 instruction = 157 * 1005 ps = 157785 ps = 157.785 ns.

b)20% of instructions have stall means that 80% of instructions have no stall = 80/100*1000 = 800 instructions, with each stage taking 157 ps.

Let us assume that the first instruction is one of the 800 instructions without stall

The number of cycles for executing 800 instructions = 6 (for first instruction) + 799 (For the remaining 800 – 1 = 799 instructions) = 805 cycles.

Total time taken by 80% of the instructions or 800 instructions = 157*805 ps.

1000 – 800 = 200 instructions, , with each stage taking 157 ps.

The number of cycles for executing 200 instructions = 200*2 (1 cycle stall for each instruction and 1 cycle for each instruction) = 400 cycles.

Total time taken by the remaining instructions or 200 instructions = 157*400 ps.

Therefore, the total time taken = 157*805 + 157*400 ps = 157*(805 + 400) ns = 157*1205 ns = 189185 ps = 189.185 ns.

2)

a)For processor P1, Clock period = Max time taken among the 5 stages = 800/5 = 160 ps.

For processor P2, Clock period = Max time taken among the 5 stages = 160 + 250 (The ALU stage takes an extra 250 ps) = 410 ps.

b)Processor P1 will execute a given code in smaller time.

Proof: Let the number of instructions for processor P1 be N, then the number of cycles for processor P1 will be N + 5 – 1 = (N + 4).

Then the number of instructions for processor P2 will be N*0.9, and the number of cycles for processor P2 will be N*0.9 + 5 – 1 = (N*0.9 + 4).

Time taken by processor P1 = 160*(N + 4) = 160*N + 640

Time taken by processor P2 = 410*(N*0.9+4) = 369*N + 1640 > 160*N + 640

c)For processor P1, Time taken by the stage which takes the maximum time for performing operations = 160 ps

The number of cycles for executing 5000 instructions = 5 (for first instruction) + 4999 (For the remaining 5000 – 1 = 4999 instructions) = 5004 cycles.

Time taken by 1000 instructions = 160 * 5004 ps = 800640 ps = 800.64 ns = 0.80064 microseconds.

For processor P2, Time taken by the stage which takes the maximum time for performing operations = 410 ps.

The number of instructions for P2 if P1 had 5000 instructions is 90/100*5000 = 4500 instructions as the number of instructions in P2 is 10% reduced as compared to P1.

The number of cycles for executing 4500 instructions = 5 (for first instruction) + 4499 (For the remaining 4500 – 1 = 4499 instructions) = 4504 cycles.

Time taken by 4500 instructions = 410 * 4504 ps = 1846640 ps = 1846.64 ns = 1.84664 microseconds.

3)

a)In the given code, line 1 modifies x14 but line 2 needs this value. Hence, we insert 2 NOPs between lines 1 and 2.

Line 3 modifies x13, while line 5 needs it. Hence, we insert an NOP anywhere between them. I have arbitrarily chosen to insert an NOP between lines 3 and 4.

Line 5 modifies x13 and line 6 needs it. Hence, we insert 2 NOPs between lines 5 and 6.

```
add x14, x12, x11
NOP_1;
NOP_2;
add x15, x14, x12
ld x13, 8(x13)
NOP_3;
ld x12, 0(x14)
and x13, x15, x13
NOP_4;
NOP_5;
ld x11, 4(x13)
sd x13, 0(x15)
```

b)With data forwarding, the NOPs inserted between lines 1 and 2; lines 3 and 4; lines 5 and 6 are no longer required.

```
add x14, x12, x11
add x15, x14, x12
ld x13, 8(x13)
ld x12, 0(x14)
and x13, x15, x13
ld x11, 4(x13)
sd x13, 0(x15)
```

4)

a)In the given code, line 1 modifies x14 while line 2 needs. Hence, we insert 2 NOPs between lines 1 and 2.

In L2, Line 3 modifies x13, while line 4 needs it. Hence, we again insert 2 NOPs between lines 3 and 4.

In L1, Line 5 modifies x13 while line 6 needs it. Hence, we again insert 2 NOPs between lines 5 and 6.

```
add x14, x12, x11
NOP_1;
NOP_2;
L2: beq x15, x14, L1
    ld x13, 8(x13)
    NOP_3;
    NOP_4;
    beq x12, x13, L2
L1: and x13, x15, x13
    NOP_5;
    NOP_6;
    ld x11, 4(x13)
    sd x13, 0(x15)
```

b)With data forwarding, the NOPs inserted between lines 5 and 6 are no longer required.

However, between lines 1 and 2; lines 3 and 4, one NOP must be there because data forwarding needs to data from MEM stage and not from EX stage for load and hence needs to wait for 1 cycle.

```
add x14, x12, x11
NOP_1
L2: beq x15, x14, L1
    ld x13, 8(x13)
    NOP_2;
    beq x12, x13, L2
L1: and x13, x15, x13
    ld x11, 4(x13)
    sd x13, 0(x15)
```